

Shanghai SHSID

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Problem

As wars started to rage in the world, the rate of amputation has gone high. More people need help on missing limbs, notably legs. The most important part of bionic leg is the ankle. Able to produce a good bionic ankle can greatly help these disabled people.

Background

The idea of a bionic ankle have existed for ages, but most bionic ankle focuses on improving only the gait pattern of the bionic ankle and shock absorption.



Figure 1: Ottobock Meridium's Bionic Leg



Figure 2: BiOM's bionic ankle

Innovation

1. Added inversion and eversion movements
2. Used a better spring system to mimic human's ankle function
3. Used ADAMS to model the kinematics for optimized lengths

Function

The bionic ankle will use two motors to mimic the movement of a normal human foot. Using the components spring system, universal joint, and motor system, the bionic foot tries to mimic the body part's function, including

1. absorbing shock,
2. providing power,
3. providing joints
4. providing supports

The Components

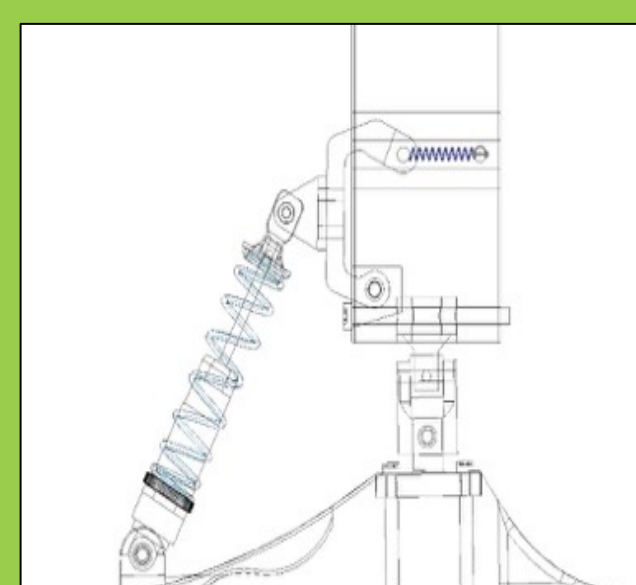


Figure 3: The spring system

Spring System: limit the foot, absorbs shock



Figure 4: The Motor system

Motor System: Drive force of the bionic foot

A Self-adaptive Multi-terrain Bionic Ankle

Solution

I am trying to make a bionic ankle that can relatively mimic a human ankle, and help disabled people to walk once more.

1. Using Solidworks, I first plan out the whole structure of hardware to check out the plan's feasibility.
2. Then, I started on building the whole project's hardware.

a) The supporting system is carefully considered to be close to an actual adult human lower leg

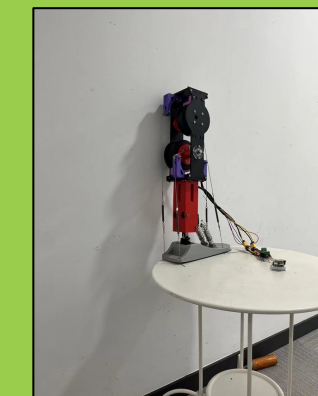


Figure 5 (left): Bionic Ankle [side view]
Figure 6 (right): Bionic Ankle [back view]



a) The motors are attached to axles, along with track rollers and flange bearings to decrease friction and attaching to supporting boards

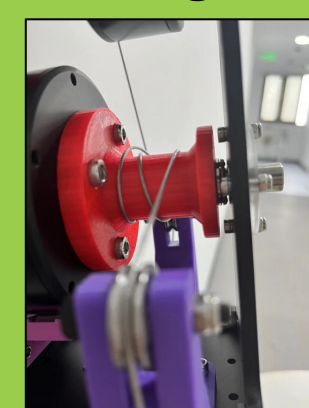
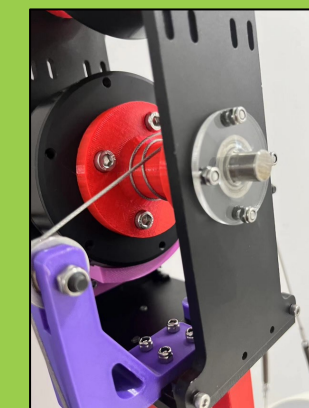


Figure 7 (left): Motor, Axle, Track rollers
Figure 8 (right): Flange Bearing



b) There are then guide wheels to guide the strings to their designated places and damping to reduce the vibration of the motors.

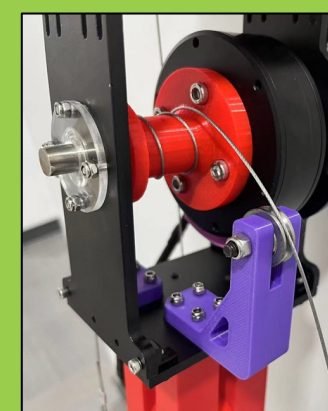
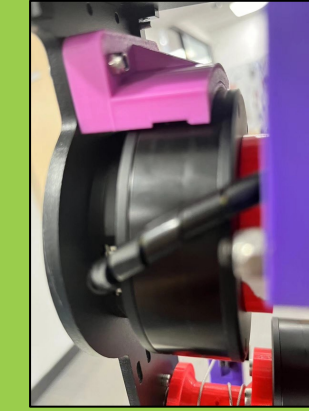


Figure 9 (left): Guide wheels
Figure 10 (right): Damping



c) The spring system contains of two parallel groups each with two springs so that the springs can apply to all directions

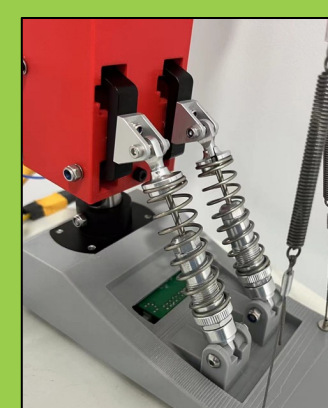
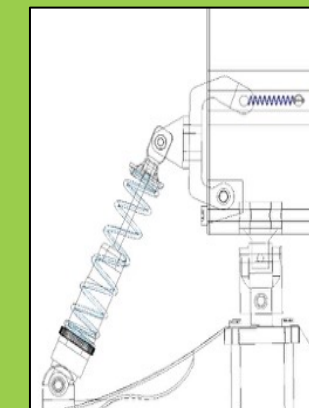


Figure 11 (left): Spring system [model]
Figure 12 (right): Spring system [transparent]



d) The universal joint is connected between the foot and the leg. The parts connecting the universal joint are specially designed to withstand power.

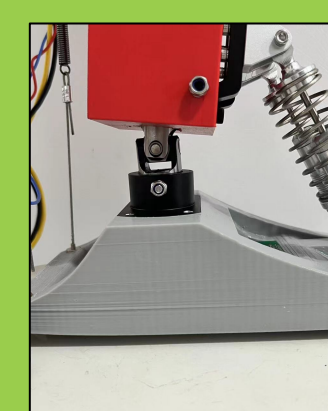
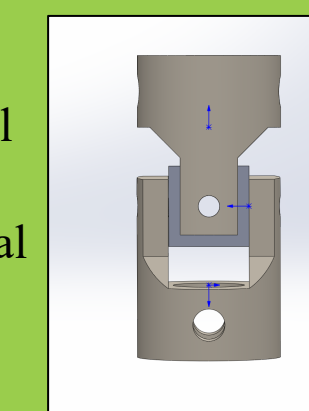


Figure 13 (left): Universal Joint [in model]
Figure 14 (right): Universal Joint



e) The strings are connected to springs to give it more stretch, and fastened tightly to the ends of the foot, wrapping around on the motors



Figure 15 (left): Springs in strings
Figure 16 (right): Foot bottom, with strings binding to it



3. I then build the Arduino setup, using esp32 as the controller, a gyroscope as a sensor, and two motors as the drive force. Esp32 can wirelessly send data to other esp32. This means the gyroscope, which is embedded into the foot, do not need to connect a wire all the way down.

Programming

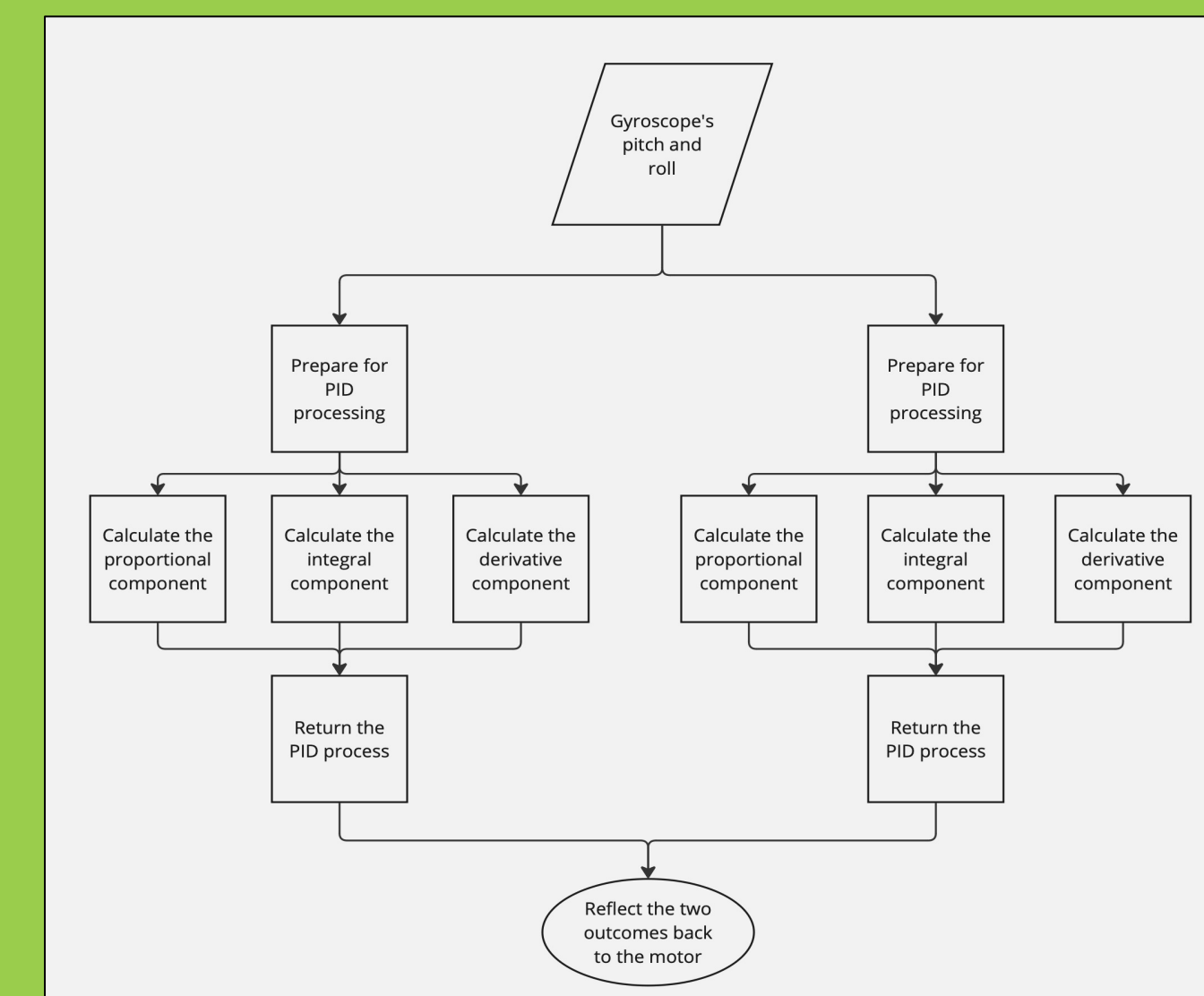


Figure 16: Software Flowchart

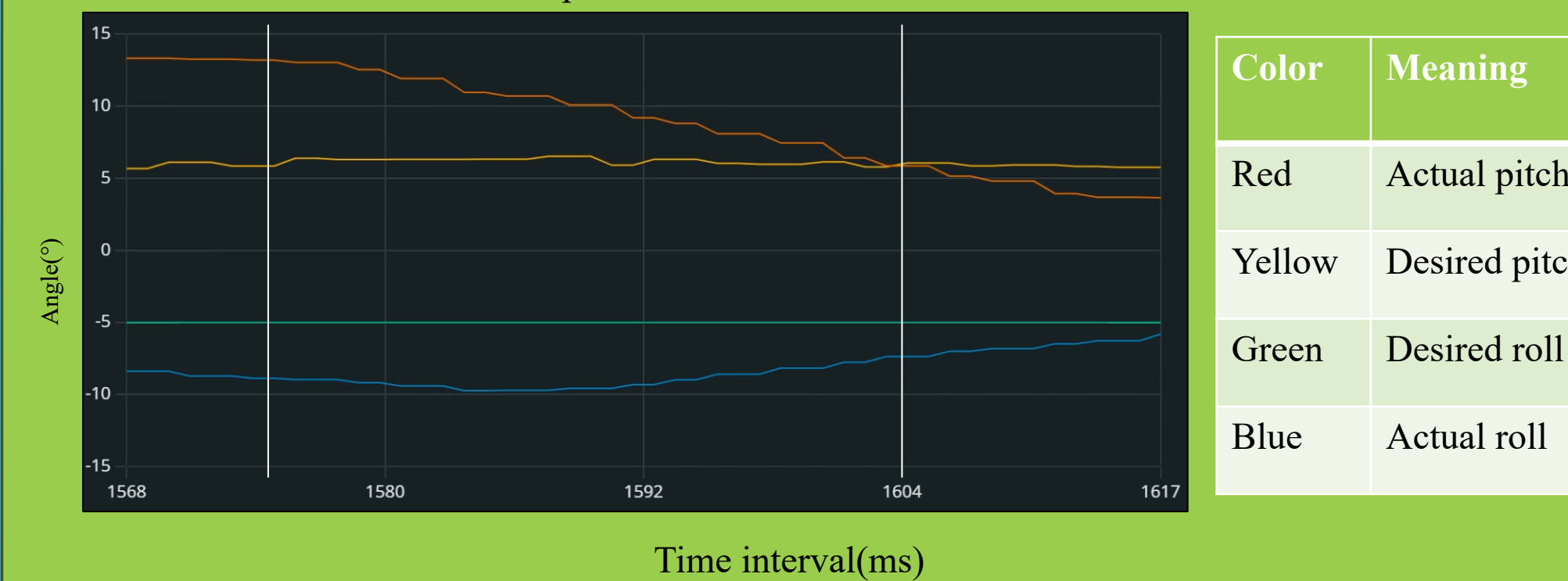
```
float PID_control(float now_data2,float set_data2)
{
    float err2 = set_data2 - now_data2;
    err_all2 = err2 + err_all2;
    if (err_all2 > 1000) err_all2 = 1000;
    if (err_all2 < -1000) err_all2 = -1000;
    float P_OUT2 = p_kp * err2;
    float I_OUT2 = p_ki * err_all2;
    float D_OUT2 = p_kd * (err2 - P.last_err);
    P.last_err = err2;
    float all_out2 = P_OUT2 + I_OUT2 + D_OUT2;
    if (all_out2 > 1000) all_out2 = 1000;
    if (all_out2 < -1000) all_out2 = -1000;
    return all_out2;
}

void PID_total(float roll, float pitch, float set_pitch, float set_roll) {
    roll_PID_angle = PID_control(roll, set_roll);
    pitch_PID_angle = PID_control(pitch, set_pitch);
    angle_increase_send(roll_PID_angle - pitch_PID_angle); delay(10);
    angle_increase_send(roll_PID_angle - pitch_PID_angle); delay(5);
}
```

Figure 17: PID controller code

The programming of the bionic ankle relies on a 2 variable PID control. The gyroscope will send its angle roll and pitch (one describes the rotation angle around the x axis, and one describes the rotation angle around the y axis). I then separately do uses PID controller on the two element. Then, combine the two outcomes and send it to the two motors, letting them increase or decrease their angle.

Graph 1: PID control outcome over time



From 1575 ms to 1600 ms, in this 25 ms time period, the PID algorithm reacted quickly to approach their desired pitch value and desired roll value.

Final Product



Figure 18: Actual Work

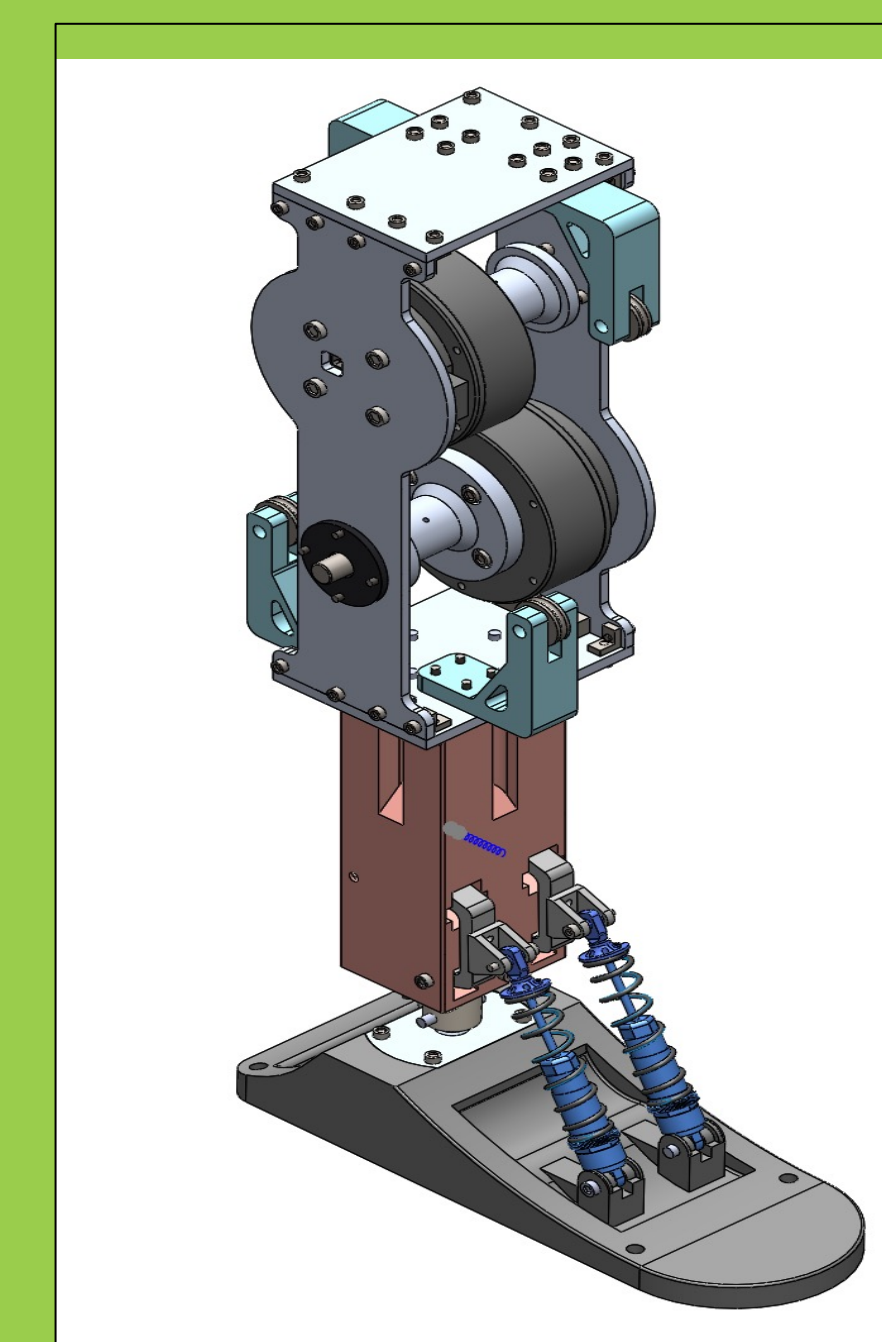


Figure 19: Solidworks Model

Experiment 1-Ankle's movement

This experiment attempts to explore how much can the foot move in the four directions: plantar flexion, dorsiflexion, inversion, and eversion.

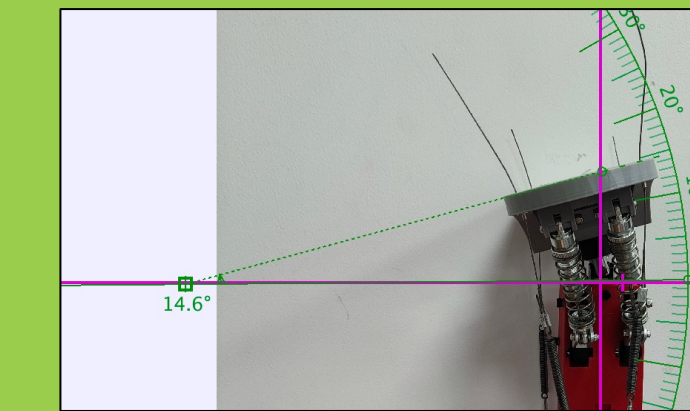


Figure 20: Angle of Eversion

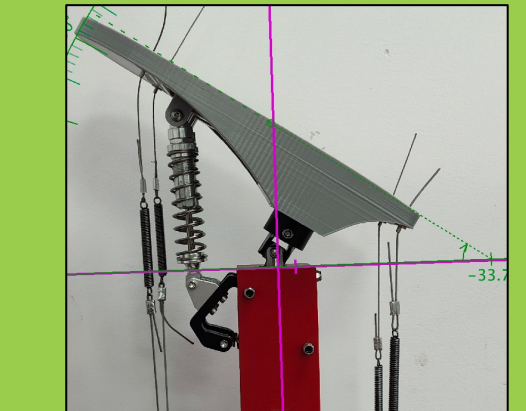


Figure 21: Angle of Planar Flexion

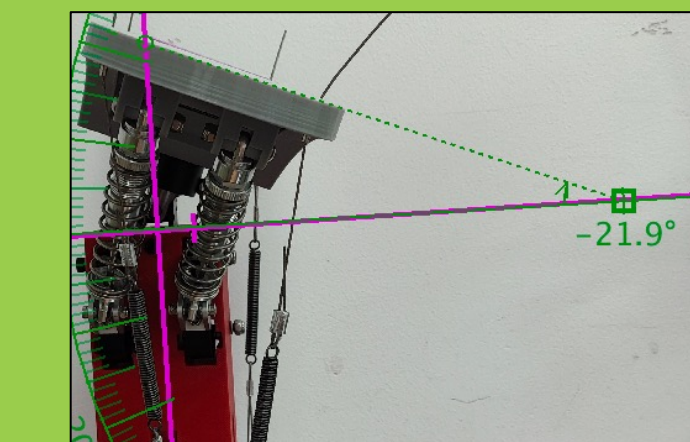


Figure 22: Angle of Inversion

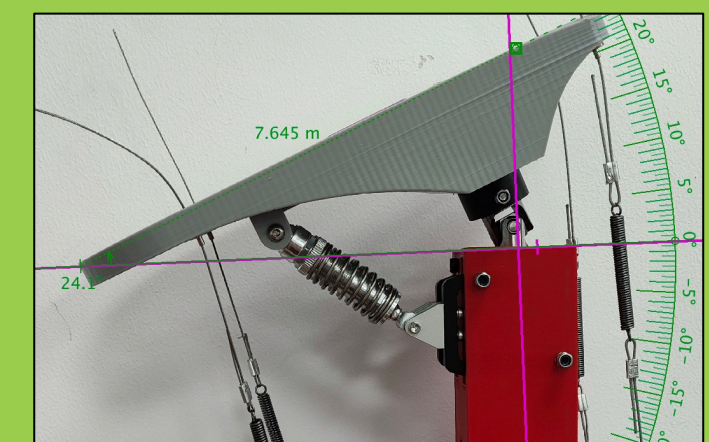


Figure 23: Angle of Dorsiflexion

Motion	Maximum angle(°)
Dorsiflexion (up)	24.1
Plantar flexion (down)	33.7
Inversion (inward)	21.9
Eversion (outward)	14.6

Table 1: The maximum angle that each movement can get

These numbers relatively matches the human's ankle movements, with the percent accuracy, except dorsiflexion, are around 26%. This means that the ankle can exceed to human ability, and even surpass it.

Experiment 2-Adaption to ground

This experiment shows the bionic ankle's adaptation to the ground. In the two diagram, the block acts like the ground. The foot will follow the ground and move according to the block.

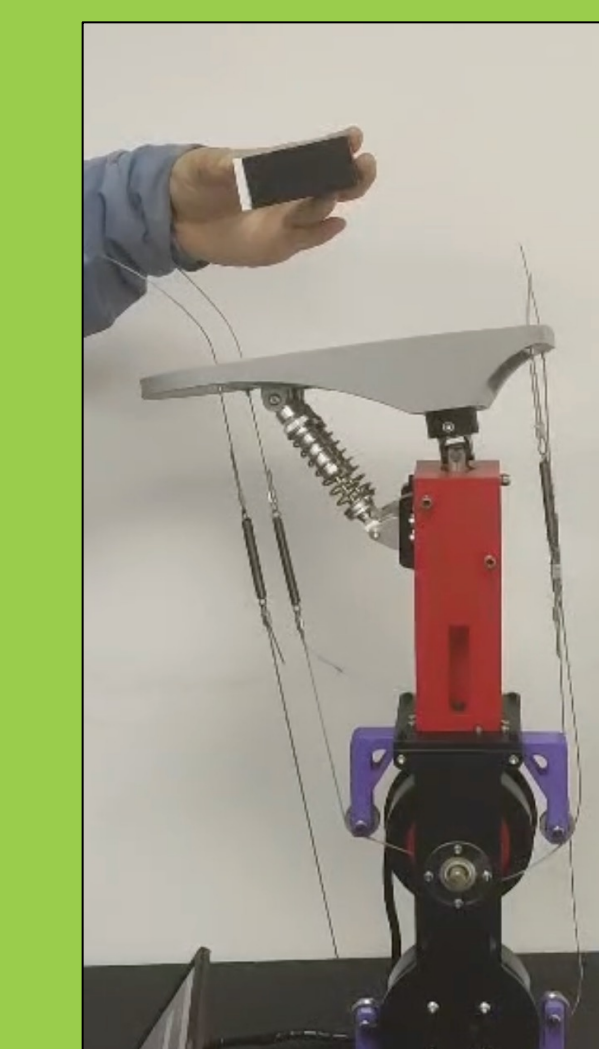


Figure 24: Before Time slice of adaptation to the ground video

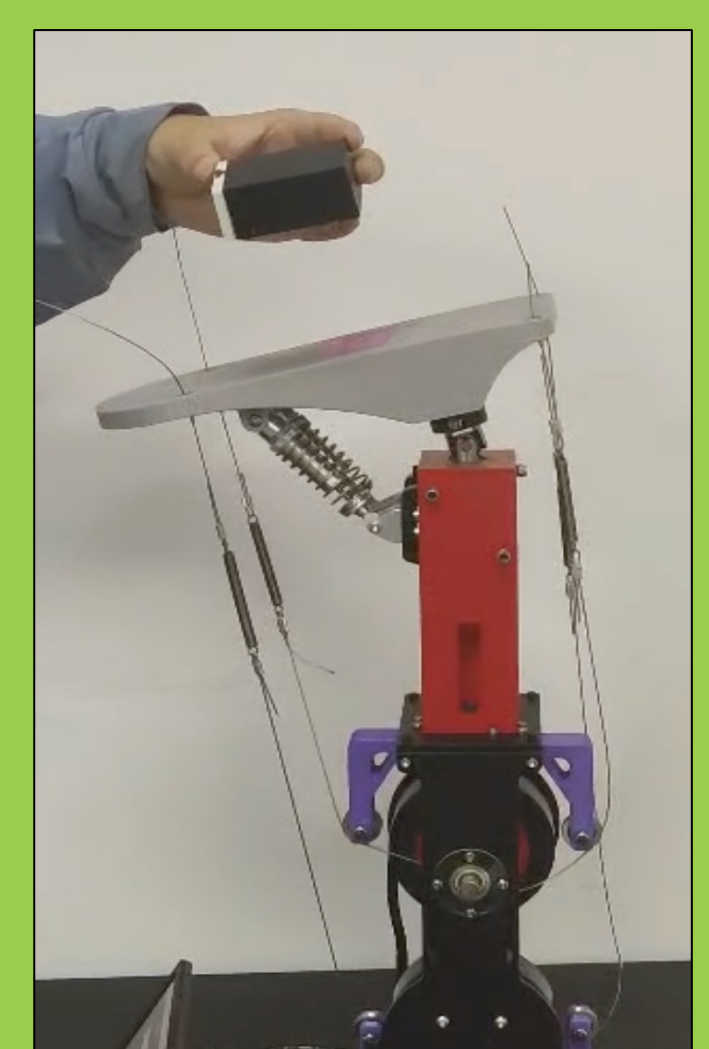
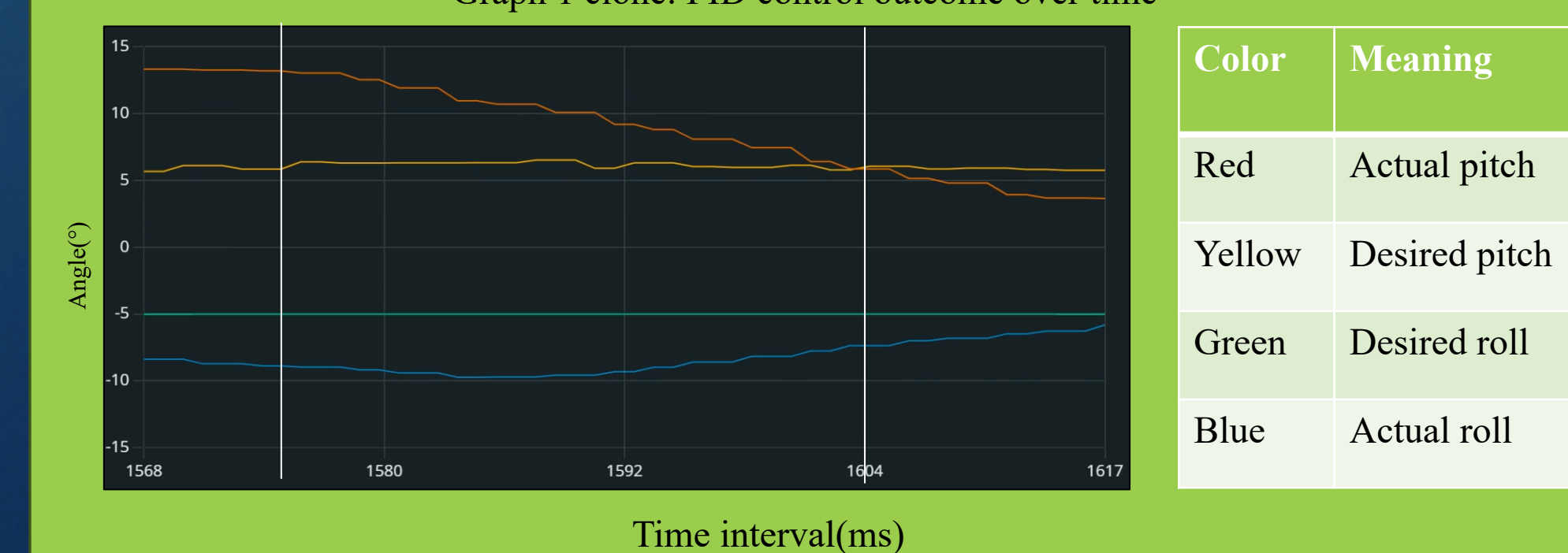


Figure 25: After Time slice of adaptation to the ground video

The below graph is the values of the experiment. As the block moves, the desired pitch and roll will change, therefore the foot's pitch and roll will move according to it.

Graph 1 clone: PID control outcome over time



This adaption to the ground can help the users tranverse complex terrains and environment.